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Vision-guided stitching systems and logic for fabricating engineered textiles with interstitched superposed wires

Abstract

Presented are automated manufacturing systems for fabricating engineered textiles, footwear and apparel formed with such engineered textiles, methods for making such engineered textiles, and memory-stored, processor-executable instructions for operating such manufacturing systems. An automated manufacturing system constructs engineered textiles from workpieces composed of superposed, unwoven wires. The system includes a movable end effector bearing a stitching head and an image capture device. The stitching head has a thread feeder and sewing needle to generate stitches. The image capture device captures images of the workpiece and outputs data indicative thereof. A system controller receives this image capture device data and locates, from the captured image of the workpiece, gaps defined between quadrangles of the superposed wires. The controller commands the end effector to sequentially move the stitching head and thereby align the sewing needle with the gaps, and commands the stitching head to insert a succession of stitches within these gaps.

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Background/Summary

CLAIM OF PRIORITY AND CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 18/066,304, which was filed on Dec. 15, 2022, is now allowed, and is a divisional of U.S. patent application Ser. No. 17/085,297, which was filed on Oct. 30, 2020, is now U.S. Pat. No. 11,555,264 B2, and claims priority to U.S. Provisional Patent Application No. 62/929,499, which was filed on Nov. 1, 2019, and is now expired. All of the foregoing patent documents are incorporated herein by reference in their respective entireties and for all purposes.

TECHNICAL FIELD

(1) The present disclosure relates generally to engineered textiles. More specifically, aspects of this disclosure relate to systems, methods, and devices for automated fabrication of engineered textiles for footwear and apparel.

BACKGROUND

(2) Articles of footwear, such as shoes, boots, slippers, sandals, and the like, are generally composed of two primary elements: an upper for securing the footwear to a user's foot; and a sole for providing subjacent support to the foot. Uppers may be fabricated from a variety of materials, including textiles, polymers, natural and synthetic leathers, etc., that are stitched or bonded together to form a shell or harness for securely receiving a foot. Many sandals and slippers, for example, have an upper with an open toe and/or open heel construction. Some designs employ an upper that is limited to a series of straps that extend over the user's instep and, optionally, around the ankle. Conversely, boot and shoe designs employ a full upper with a closed toe and heel construction that encases the foot. An ankle opening through a rear quarter portion of the upper provides access to the footwear's interior, facilitating entry and removal of the foot into and from the upper. A shoelace or strap system may be utilized to secure the foot within the upper.

(3) A sole structure is mounted to the underside of the upper, positioned between the user's foot and the ground. In many articles of footwear, including athletic shoes and boots, the sole structure is a layered construction that generally incorporates a comfort-enhancing insole, an impact-mitigating midsole, and a surface-contacting outsole. The insole, which may be located partially or entirely within the upper, is a thin and compressible member that provides a contact surface for the underside "plantar" region of the user's foot. By comparison, the midsole is mounted underneath the insole, forming a middle layer of the sole structure. In addition to attenuating ground reaction forces, the midsole may help to control foot motion and impart enhanced stability. Secured underneath the midsole is an outsole that forms the ground-contacting portion of the footwear. The outsole is usually fashioned from a durable, waterproof material that includes tread patterns engineered to improve traction.

(4) Footwear that employ a full upper with a closed toe/heel design will conventionally take on multilayer constructions that are formed by joining together a variety of cutout sheet material elements. These sheet elements may be selected to impart wear-resistance, moisture-control, stretchability, flexibility, air-permeability, comfort, etc., to different areas of the upper. To fabricate the upper, the individual elements are first cut from sheet stock to desired shape, and then joined together through stitching, adhesive bonding, or other suitable joining technique. The sheet elements are often joined in an overlapping or layered configuration to impart multiple properties to individual areas. As the number and type of sheet elements incorporated into the upper increases, the time and expense associated with transporting, stocking, cutting, and joining the elements increases proportionately. Waste material from these manufacturing processes also accumulates to a greater degree with the increase in the number and type of sheet elements incorporated into an upper. Moreover, recycling an article of footwear becomes increasingly more difficult for uppers manufactured from a large number of individual sheet elements.

SUMMARY

(5) Presented herein are automated manufacturing systems with attendant control logic for

fabricating engineered textiles, footwear and apparel formed, in whole or in part, from such engineered textiles, methods for making such engineered textiles, and memory-stored, processor-executable instructions for operating such manufacturing systems. By way of example, and not limitation, there are disclosed engineered textiles composed of superposed, unwoven wires that are interconnected, e.g., via an array of interleaved stitch seams or other joining techniques. The resultant textile does not require and, thus, may eliminate a subjacent support scrim or layer of fabric. In contrast to conventional designs, at least some of the disclosed engineered textiles are neither woven nor knitted; rather, individual strands may extend in two, three, or more directions and joined to one another at multiple predefined locations, e.g., via bonding agents, fasteners, adhesives, welding, etc.

(6) During assembly, the superposed wires may be wound around and retained in tension by the posts of a workpiece frame (or “jig”) to align the wires in an intercrossed pattern. One set of mutually parallel wire windings is elongated in a first direction, e.g., aligned with a first pre-defined load path, and another set of mutually parallel wire windings is elongated in a second direction e.g., aligned with a second pre-defined load path that is angled with respect to the first direction. Third, fourth, fifth, etc., sets may each be elongated in a respective direction that is distinct from the other sets. The first set of wire windings may be laid across and abut the second set of wire windings without interlacing the two sets of windings. To maintain a desired shape of the engineered textile, while permitting inter-wire movement, the two sets of wire windings are mechanically joined by first (top) and second (bobbin) threads lockstitched together in the gaps between the superposed wires. The lockstitched threads may be arranged in a matrix of orthogonal rows and columns, which interleave with and abut against the wires. Alternatively, the wire windings may be joined via adhesives, fasteners, fusing, etc.

(7) Assembling the above-mentioned engineered textiles may be complicated by a variety of considerations, including retaining the superposed wires in tension while joining them together, and joining the wires in a manner that allows for wire-on-wire translation while preventing the textile from losing shape or becoming tangled once removed from the jig. Other complications may include preventing wire movement during stitching, locating a central gap defined between each quadrangle of crisscrossed wires, and precision lockstitching together the top and bobbin threads in these central gaps, etc. To address any one or more or all of the foregoing issues, an automated manufacturing system is presented that employs a jig for maintaining wire positioning and tension, and a vision or laser-guided stitching head for precision locating of interwire gaps and interconnecting the superposed wires. For some implementations, the automated manufacturing system may utilize a precision positioning apparatus with a laser-based alignment sensor to hold, orient, and dynamically position the jig and, thus, the superposed wires. Additionally, or alternatively, a stitching end effector with a stitching head and a high-precision digital camera is mounted to a robot arm or carriage for controller-automated, vision guided stitching of the superposed wires.

(8) Aspects of this disclosure are directed to controller-regulated, vision-guided stitching systems for assembling engineered textiles. In an example, an automated manufacturing system is presented for constructing an engineered textile from a workpiece composed of superposed wires. By way of contrast to existing sewing systems that are delimited to stitching together woven fabrics cutouts, polymeric sheets, natural and synthetic leather panels, etc., this automated manufacturing system is generally intended to mechanically connect an unwoven, intercrossed array of wire windings. These windings may be formed from any suitable natural or synthetic material, including extruded elastic and inelastic polymers, braided fibers, combinations thereof, and the like. The automated manufacturing system includes a movable end effector, such as a pneumatic articulating robot arm or a motor-driven carriage. A stitching head, which is mounted to the movable end effector, includes one or more thread feeders and a sewing needle that cooperatively generate stitches. Also mounted to the movable end effector is an image capture device that captures images of the

workpiece and outputs data indicative thereof.

(9) Continuing with the discussion of the above example, the automated manufacturing system also includes a resident or remote system controller, which may be embodied as an electronic control unit or a network of distributed controllers or control modules, for regulating operation of one or more resident processing systems. The system controller is wired or wirelessly connected to the movable end effector, stitching head, and image capture device. This controller is programmed to receive, from the image capture device, the data indicative of the captured image of the workpiece, and locate, from the captured image, multiple gaps each defined between a quadrangle of the superposed wires. Once the interwire gaps are located, the system controller transmits one or more command signals to the movable end effector to sequentially move the stitching head across the workpiece and thereby align the sewing needle with each of the identified gaps. The system controller concurrently transmits one or more command signals to the stitching head to insert a succession of stitches within the gaps between the superposed wires.

(10) Other aspects of this disclosure are directed to footwear, apparel, sporting goods, and other consumer products fabricated with any of the disclosed engineered textiles. As an example, an article of footwear is presented that includes an upper designed to receive and attach to a foot of a user, and a sole structure that is attached to the upper and designed to support thereon the user's foot. The upper is fabricated, in whole or in part, from an engineered textile and, thus, includes one or more upper segments that are manufactured from engineered textiles. The engineered textile may include a first set of mutually parallel wire windings elongated in a first direction, and a second set of mutually parallel wire windings elongated in a second direction that is distinct from (e.g., obliquely angled or substantially orthogonal to) the first direction. The first and second sets of wire windings are superposed such that the first set abuts the second set in an unwoven, intercrossed pattern defining an array of quadrangles each having a central gap. First (top) and second (bobbin) threads are elongated in a third and, optionally, a fourth direction that are respectively parallel with respect to the first and second directions. In another embodiment, the first and second threads may define a third direction that is oblique and/or orthogonal to the first and second directions. These two threads are lockstitched together with a respective lockstitch disposed in each central gap between intercrossed wire windings.

(11) Additional aspects of the present disclosure are directed to techniques, algorithms, and logic for operating any of the disclosed systems or for manufacturing any of the disclosed engineered textiles. For instance, non-transitory, computer-readable media (CRM) are presented that store instructions executable by one or more processors of a system controller of an automated manufacturing system. These instructions cause the automated manufacturing system to perform a set of system operations, including receiving, from an image capture device mounted to a movable end effector, data indicative of a captured image of a workpiece. The workpiece is composed of multiple unwoven, superposed wire windings, e.g., aligned in a crisscross pattern. The movable end effector also has mounted thereto a stitching head with a thread feeder and a sewing needle that are cooperatively configured to generate stitches. The stored instructions also cause the system to locate, from the captured image of the workpiece, multiple gaps defined between individual quadrangles of the superposed, intercrossed wires. One or more command signals are sent to the movable end effector to sequentially move the stitching head and thereby align the sewing needle with each of the gaps. In addition, one or more command signals are sent to the stitching head to insert a succession of stitches within the gaps between the superposed wires.

(12) Additional aspects of this disclosure are directed to methods for manufacturing any of the disclosed engineered textiles and methods for controlling any of the disclosed systems and devices. In an example, a method is presented for operating an automated manufacturing system for constructing an engineered textile from a workpiece composed of superposed wires. This representative method includes, in any order and in any combination with any of the above or below disclosed features and options: receiving, via a system controller from an image capture

device mounted to a movable end effector, data indicative of a captured image of a workpiece, the movable end effector having mounted thereto a stitching head with a thread feeder and a sewing needle cooperatively configured to generate stitches; locating, via the system controller from the captured image of the workpiece, multiple gaps each defined between a quadrangle of the superposed wires; commanding, via the system controller, the movable end effector to sequentially move the stitching head and thereby align the sewing needle with each of the gaps; and commanding, via the system controller, the stitching head to insert a succession of stitches within the gaps between the superposed wires.

(13) For any of the disclosed manufacturing systems, methods, and CRM, the system controller may identify, within the captured image of the workpiece, respective sets of intersecting points (e.g., four points per set) of the superposed wires defining the quadrangles. The controller then determines, within each respective set, a center of a respective diagonal line segment connecting an opposing pair of the intersecting points. In this instance, locating the quadrangle gaps includes designating the center of the diagonal line segment of each set of intersecting points as one of the gaps. As another option, the system controller may identify, within the captured workpiece image, a respective estimated centerline for each superposed wire, and construct the quadrangles of the superposed wires from these estimated centerlines. In this instance, locating the gaps includes designating a central region within each of the quadrangles between the estimated centerlines as one of the gaps. Optionally, the system controller may identify, within the captured image, at least two intersecting points of the superposed wires each defining a respective corner of a quadrangle, and determine a central region for each quadrangle at a calibrated angle from a line segment connecting the two respective corners and a calibrated distance from one of the corners. In this instance, locating the gaps includes designating the central region of each quadrangle as one of the gaps.

(14) For any of the disclosed manufacturing systems, methods, and CRM, the system controller may derive, calculate, retrieve, or look-up (hereinafter “determine”) path plan data for the stitching head to insert the succession of stitches within the gaps between the superposed wires. The path plan data includes a path origin, a path destination, and a stitch route for traversing the stitching head from the origin to the destination. As part of this procedure, the system controller may optionally generate a trace of the stitch route, determine start and end positions within the captured image of the workpiece, and superimpose the stitch route trace onto the captured image with the origin overlapping the start position and the destination overlapping the end position. the system controller may then determine multiple calibrated alignment points on the stitch route, determine a respective displacement, if any, between each calibrated alignment point and a respective alignment location in the workpiece image, and determine a respective trace correction to offset each respective displacement.

(15) For any of the disclosed engineered textiles, CRM, manufacturing systems and methods, the manufacturing system may be equipped with a workpiece frame that is structurally configured to retain the superposed wires in a tensioned, crisscrossed pattern. The workpiece frame may be fabricated with multiple adjoining casing walls that define therebetween an inner frame space across which the workpiece is stretched. A series of posts project, e.g., substantially orthogonally, from the casing walls, with the posts spaced from one another along the perimeter of the inner frame space. In this instance, the wires are wound around and suspended from the posts.

(16) For any of the disclosed manufacturing systems, methods, and CRM, the manufacturing system may be equipped with one or more sensors that track, in real-time, the movement of the stitching head relative to a calibrated origin position. The system controller may optionally receive one or more sensor signals from one or more position sensors indicative of real-time positions of the stitching head. From the received sensor signal(s) and captured workpiece image(s), the controller may determine an estimated distance between each real-time position of the stitching head and a next adjacent one of the gaps. In this instance, commanding the movable end effector to

sequentially move the stitching head includes estimating multiple desired trajectories each based on the estimated distance between a real-time position of the stitching head and a respective next adjacent gap. The system controller may concurrently determine, one-at-a-time in real-time from the received sensor signal(s) and captured workpiece image(s), a respective next adjacent gap closest to each real-time position of the stitching head.

(17) For any of the disclosed engineered textiles, CRM, manufacturing systems and methods, the stitching head may be equipped with a needle receiver that is operable to reciprocally translate the sewing needle, a bobbin case that is operable to feed bobbin thread, and a shuttle hook that is operable to create a lockstitch between the bobbin thread and a top thread fed from the thread feeder. For some system configurations, the movable end effector is comprised of a support frame attached to a robot arm. Alternatively, the movable end effector is comprised of a support carriage attached to a slide track frame.

(18) The above summary does not represent every embodiment or every aspect of the present disclosure. Rather, the foregoing summary merely provides an exemplification of some of the novel concepts and features set forth herein. The above features and advantages, and other features and attendant advantages of this disclosure, will be readily apparent from the following detailed description of illustrated examples and representative modes for carrying out the present disclosure when taken in connection with the accompanying drawings and the appended claims. Moreover, this disclosure expressly includes any and all combinations and subcombinations of the elements and features presented above and below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a lateral, side-view illustration of a representative article of footwear with one or more engineered textile upper segments in accordance with aspects of the present disclosure.

(2) FIG. 1B is a bottom-view illustration of the representative article of footwear of FIG. 1A.

(3) FIG. 2 is an elevated, perspective-view illustration of a representative athletic shoe with an engineered textile upper in accord with aspects of the disclosed concepts.

(4) FIG. 3 is a plan-view illustration of a representative workpiece frame (“jig”) retaining therein a representative workpiece composed of unwoven, superposed wires tensioned in a crisscrossed pattern.

(5) FIG. 4 is a schematic illustration of a representative automated manufacturing system for constructing engineered textiles from workpieces composed of unwoven, superposed wires in accord with aspects of the disclosed concepts.

(6) FIG. 5 is a flowchart illustrating a representative control algorithm with logic for controlling operation of an automated manufacturing system, which may correspond to memory-stored instructions executed by a resident or remote controller, control-logic circuitry, programmable electronic control unit, or other integrated circuit (IC) device or a network of IC devices in accord with aspects of the disclosed concepts.

(7) The present disclosure is amenable to various modifications and alternative forms, and some representative embodiments are shown by way of example in the drawings and will be described in detail herein. It should be understood, however, that the novel aspects of this disclosure are not limited to the particular forms illustrated in the above-enumerated drawings. Rather, the disclosure is to cover all modifications, equivalents, combinations, subcombinations, permutations, groupings, and alternatives falling within the scope of this disclosure as encompassed by the appended claims.

DETAILED DESCRIPTION

(8) Aspects of the present disclosure broadly relate to an article of footwear formed using one or more non-woven engineered textiles, and manufacturing methods for creating such textiles. In

general, the engineered textiles of the present disclosure are comprised of a plurality of tensile strands that may be selectively positioned and oriented along certain specified load paths such that the textile may predictably respond during certain functional activities. Because the textile is formed without a weave, material integrity may devolve into a spaghetti-like mess of strands absent some manner of joining adjacent layers. As such, the present disclosure broadly relates to manners of adaptively joining adjacent layers of obliquely angled tensile strands absent a weave. As described, it is preferred that the manner of joining permits some degree of relative wire movement, as opposed to rigidly locking all strands into a rigid alignment. This local movement may allow the textile to move and respond any flexure of the wearer's body throughout the functional activity while still maintaining overall material integrity. While the present disclosure primarily describes joining via a lock stitch at wire intersection points, such should be regarded as merely an example unless so limited by the claims.

(9) This disclosure is susceptible of embodiment in many different forms. Representative examples of the disclosure are shown in the drawings and will be described in detail herein with the understanding that these representative examples are provided as an exemplification of the disclosed principles, not limitations of the broad aspects of the disclosure. To that extent, elements and limitations that are described in the Abstract, Technical Field, Background, Summary, and Detailed Description sections, but not explicitly set forth in the claims, should not be incorporated into the claims, singly or collectively, by implication, inference or otherwise.

(10) For purposes of the present detailed description, unless specifically disclaimed: the singular includes the plural and vice versa; the words “and” and “or” shall be both conjunctive and disjunctive; the words “any” and “all” shall both mean “any and all”; and the words “including”, “comprising”, “having”, “containing”, and the like shall each mean “including without limitation.” Moreover, words of approximation, such as “about,” “almost”, “generally”, “substantially”, “approximately”, and the like, may be used herein in the sense of “at, near, or nearly at,” or “within 0-5% of,” or “within acceptable manufacturing tolerances”, or any logical combination thereof, for example. Lastly, directional adjectives and adverbs, such as fore, aft, medial, lateral, proximal, distal, vertical, horizontal, front, back, left, right, etc., may be with respect to an article of footwear when worn on a user's foot and operatively oriented with a ground-engaging bottom surface of the sole structure seated on a flat surface, for example.

(11) Referring now to the drawings, wherein like reference numbers refer to like features throughout the several views, there is shown in FIG. 1A a representative article of footwear, which is designated generally at **10** and portrayed herein for purposes of discussion as an athletic shoe or “sneaker”. The illustrated article of footwear **10**—also referred to herein as “footwear” or “shoe” for brevity—is an exemplary application with which novel aspects and features of this disclosure may be practiced. In the same vein, implementation of the present concepts by the illustrated automated manufacturing system should also be appreciated as a representative implementation of the disclosed concepts. It will therefore be understood that aspects of this disclosure may be integrated into other footwear designs, may be incorporated into any logically relevant type of consumer product, and may be carried out by other automated manufacturing system architectures. As used herein, the terms “shoe” and “footwear”, including permutations thereof, may be used interchangeably and synonymously to reference any suitable type of garment worn on a human foot. Lastly, features presented in the drawings are not necessarily to scale and are provided purely for instructional purposes. Thus, the specific and relative dimensions shown in the drawings are not to be construed as limiting.

(12) The representative article of footwear **10** is generally depicted in FIGS. 1A and 1B as a bipartite construction that is primarily composed of a foot-receiving upper **12** mounted on top of a subjacent sole structure **14**. For ease of reference, footwear **10** may be divided into three anatomical regions: a forefoot region R.sub.FF, a midfoot region R.sub.MF, and a hindfoot (heel) region R.sub.HF, as shown in FIG. 1. Footwear **10** may also be divided along a vertical plane into a

lateral segment S.sub.LA—a distal half of the shoe **10** farthest from the sagittal plane of the human body—and a medial segment S.sub.ME—a proximal half of the shoe **10** closest to the sagittal plane of the human body. In accordance with recognized anatomical classification, the forefoot region R.sub.FF is located at the front of the footwear **10** and generally corresponds with the phalanges (toes), metatarsals, and any interconnecting joints thereof. Interposed between the forefoot and hindfoot regions R.sub.FF and R.sub.HF is the midfoot region R.sub.MF, which generally corresponds with the cuneiform, navicular and cuboid bones (i.e., the arch area of the foot). Hindfoot region R.sub.HF, in contrast, is located at the rear of the footwear **10** and generally corresponds with the talus (ankle) and calcaneus (heel) bones. Both lateral and medial segments S.sub.LA and S.sub.ME of the footwear **10** extend through all three anatomical regions R.sub.FF, R.sub.MF, R.sub.HF, and each corresponds to a respective transverse side of the footwear **10**. While only a single shoe **10** for a left foot of a user is shown in FIGS. **1A** and **1B**, a mirrored, substantially identical counterpart for a left foot of a user may be provided. Recognizably, the shape, size, material composition, and method of manufacture of the shoe **10** may be varied, singly or collectively, to accommodate practically any conventional or nonconventional footwear application.

(13) With reference again to FIG. **1A**, the upper **12** is depicted as having a shell-like closed toe and heel configuration for encasing a human foot. Upper **12** of FIG. **1A** is generally defined by three adjoining sections, namely a toe box **12A**, a vamp **12B** and a rear quarter **12C**. The toe box **12A** is shown as a rounded forward tip of the upper **12** that extends from distal to proximal phalanges to cover and protect the user's toes. By comparison, the vamp **12B** is an arched midsection of the upper **12** that is located aft of the toe box **12A** and extends from the metatarsals to the cuboid. As shown, the vamp **12B** also provides a series of lace eyelets **16** and a shoe tongue **18**. Positioned aft of the vamp **12B** is a rear quarter **12C** that extends from the transverse tarsal joint to wrap around the calcaneus bone, and includes the rear end and rear sides of the upper **12**. While portrayed in the drawings as comprising three primary segments, the upper **12** may be fabricated as a single-piece construction or may be composed of any number of segments, including a toe shield, heel cap **30**, ankle cuff, interior liner, etc. For sandal and slipper applications, the upper **12** may take on an open toe or open heel configuration, or may be replaced with a single strap or multiple interconnected straps.

(14) The upper **12** portion of the footwear **10** may be fabricated from any one or combination of a variety of materials, such as textiles, engineered foams, polymers, natural and synthetic leathers, etc. Individual segments of the upper **12**, once assembled or cut to shape and size, may be stitched, adhesively bonded, fastened, welded or otherwise joined together to form an interior void for comfortably receiving a foot. The individual material elements of the upper **12** may be selected and located with respect to the footwear **10** in order to impart desired properties of durability, air-permeability, wear-resistance, flexibility, appearance, and comfort, for example. An ankle opening **15** in the rear quarter **12C** of the upper **12** provides access to the interior of the shoe **10**. A shoelace **20**, strap, buckle, or other commercially available mechanism may be utilized to modify the girth of the upper **12** to more securely retain the foot within the interior of the shoe **10** as well as to facilitate entry and removal of the foot from the upper **12**. Shoelace **20** may be threaded through a series of eyelets **16** in or attached to the upper **12**; the tongue **18** may extend between the lace **20** and the interior void of the upper **12**.

(15) Sole structure **14** is rigidly secured to the upper **12** such that the sole structure **14** extends between the upper **12** and a support surface upon which a user stands. In effect, the sole structure **14** functions as an intermediate support platform that separates and protects the user's foot from the ground. In addition to attenuating ground reaction forces and providing cushioning for the foot, sole structure **14** of FIGS. **1A** and **1B** may provide traction, impart stability, and help to limit various foot motions, such as inadvertent foot inversion and eversion. It is envisioned that the sole structure **14** may be attached to the upper **12** via any presently available or hereinafter developed

suitable means. For at least some applications, the upper **12** may be coupled directly to the midsole **24** and, thus, lack a direct coupling to either the insole **22** or the outsole **26**. By way of non-limiting example, the upper **12** may be adhesively attached to only an inside periphery of a midsole sidewall **21**, e.g., secured with a 10 mm bonding allowance via priming, cementing, and pressing.

(16) In accordance with the illustrated example, the sole structure **14** is fabricated as a sandwich structure with a foot-contacting insole **22** (FIG. 1A), an intermediate midsole **24**, and a bottom-most outsole **26**. Alternative sole structure configurations may be fabricated with greater or fewer than three layers. Insole **22** is located within an interior void of the footwear **10**, operatively located at a lower portion of the upper **12**, such that the insole **22** abuts a plantar surface of the foot. Underneath the insole **22** is a midsole **24** that incorporates one or more materials or embedded elements that enhance the comfort, performance, and/or ground-reaction-force attenuation properties of footwear **10**. These elements and materials may include, individually or in any combination, a polymer foam material, such as polyurethane or ethyl vinyl acetate (EVA), filler materials, moderators, air-filled bladders, plates, lasting elements, or motion control members. Outsole **26** is located underneath the midsole **24**, defining only some or all of the bottom-most, ground-engaging portion of the footwear **10**. The outsole **26** may be formed from a natural or synthetic rubber material that provides a durable and wear-resistant surface for contacting the ground. In addition, the outsole **26** may be contoured and textured to enhance the traction (i.e., friction) properties between footwear **10** and the underlying support surface.

(17) With reference now to FIG. 2, there is shown another representative article of footwear, which is designated generally at **110** and portrayed herein for purposes of discussion as an athletic shoe of the basketball type. Although differing in appearance, the athletic shoe **110** of FIG. 2 may take on any of the features, options, and alternatives described above with respect to the footwear **10** presented in FIGS. 1A-1B, and vice versa. For instance, the athletic shoe **110** of FIG. 2 includes a foot-securing upper **112** that is seated on top of a foot-supporting sole structure **114**. Athletic shoe **110** is also assembled with an elongated tongue **118** that extends between a shoelace **120** and an interior foot-receiving void of the upper **112**. By way of comparison, the footwear **10** of FIG. 1A is assembled with one or more discrete upper sections, such as midfoot vamp section **32**, each fabricated from an engineered textile formed from superposed, interstitched wires. Footwear **110** of FIG. 2, on the other hand, is assembled with an upper **112** having an outer surface that is fabricated almost entirely from the engineered textile material **132**. In particular, the engineered textile **132** surface of upper **112** extends from a forward edges of the toe box **112A**, through both sides of the vamp section **112B**, and around the rear quarter **112C**. An optional translucent scrim layer **134** extends across and covers select sections of the upper's **112** engineered textile **132** surface, providing structural reinforcement to those select segments of the upper **112**.

(18) Inset within FIG. 2 are enlarged illustrations of the engineered textile **132** used to fabricate the exterior surface of the footwear upper **112**. In accord with the illustrated example, the engineered textile **132** generally comprises or, for at least some implementations, consists essentially of two sets of wire windings **140** and **142** that are interconnected via an array of interleaved stitch seams formed from opposing threads **144** and **146**. The windings of the first set of wire windings **140** are substantially parallel to one another and all elongated in a first direction **D1**, e.g., corresponding to a first pre-defined load path. In the same vein, the windings of the second set of wire windings **140** are substantially parallel to one another and all elongated in a second direction **D2**, which may be generally orthogonal to the first direction **D1** and may correspond to a second pre-defined load path. While shown arranged in a square array of perpendicular rows and columns, it is envisioned that the first set of wire windings **140** may be obliquely angled with respect to the second set of wire windings **142**. These windings **140**, **142** may be formed from an organic or inorganic material, including extruded elastic polymers, braided elastic fibers, inelastic polymer fibers, combinations thereof, and the like. In some embodiments, the strands **140**, **142** may comprise one or more of an aliphatic or semi-aromatic polyamide fiber, such as PA6, PA66, an aromatic polyester fiber, such as

VECTRAN®—manufactured by Kuraray Co., Ltd, an aramid fiber, such as KEVLAR®—manufactured by DuPont de Nemours, Inc, a polypropylene fiber or a high modulus polyethylene fiber. It may be desirable, depending on desired application, that the diameters of the windings **140**, **142** be at least 50-75% larger or, in some embodiments, at least 100-200% larger or, in some embodiments, 3-times to 4-times larger than the diameters of the stitching threads **144**, **146**.

(19) To help ensure that the wire windings **140**, **142** are assembled in a manner that allows for relative wire movement and/or wire-on-wire translation, the first set of wire windings **140** are located on top of the second set of wire windings **142** in an abutting, non-woven manner. Rather than interlace the wire windings **140**, **142** in an alternating over-under composition, as might be seen in a conventional woven textile sheet, the first set of wire windings **140** lays across an upper face of the second set of wire windings **142** in an unwoven, intercrossed pattern. In so doing, first wire windings **140** may translate and/or stretch in the first direction **D1**, and second wire windings **142** may translate and/or stretch in the second direction **D2** independent of or contemporaneous with the translating/stretching first wire windings **140**. Intersecting the wire windings **140**, **142** in a crisscross arrangement defines an array of quadrangles, four of which are shown hidden in the upper righthand inset view of FIG. **2** and designated generally as **148**. These quadrangles **148** are portrayed in the Figures as right-rectangular polygons; nevertheless, each quadrangle **148** may take on other shapes and sizes, which may be similar to or distinct from the quadrilateral shapes of the other quadrangles **148**. At the midpoint of each quadrangle **148** is a central through hole or “interwire gap” **150**.

(20) The stacked wires **140**, **142** are mechanically joined in a manner that maintains a desired perimeter shape of the assembled engineered textile **132**, yet does not impede the above-described wire-on-wire movement. According to the illustrated example, a first thread **144**—known in sewing parlance as the “top thread”—is interlaced with a second thread **146**—known as the “bobbin thread”—through an automated stitching process in order to form an assortment of substantially linear stitch seams that are interleaved with and bind together the crisscrossed sets of wire windings **140**, **142**. The upper righthand inset view of FIG. **2** depicts one set of the linear stitch seams elongated in a third direction **D3** that is substantially parallel with respect to the first direction **D1** of the first wire set **140**. An optional second set of the linear stitch seams is elongated in a fourth direction **D4** that is substantially parallel with respect to the second direction **D2** of the second wire set **142**. Fashioning the first set of stitch seams may be achieved by drawing the first and second threads **144**, **146** in the third direction **D3** and sequentially lockstitching them together in the central gaps **150** between each grouping of intercrossed wires. Likewise, the second set of stitch seams may be fashioned by drawing the first and second threads **144**, **146** in the fourth direction **D4** and sequentially lockstitching them together in the central gaps **150** between each grouping of intercrossed wires. This process is systematically repeated along parallel trajectories aligned with the third and fourth directions **D3** and **D4** until a desired number of linear stitch seams is achieved. As seen in the lower left-hand inset view of FIG. **2**, the interlocking threads **144**, **146** may be drawn in directions that are oblique with respect to the intercrossed wires.

(21) It should be recognized that the structural integrity of the engineered textile **132** may be optimized by placing a lockstitch inside each interwire gap **150**; however, it is within the scope of this disclosure to place a lockstitch in every other gap **150** or in only selected ones of the gaps **150**, e.g., using controller-automated, vision-guided stitching techniques. Optimized structural integrity may be further optimized by positioning a stitch seam between every pair of neighboring, parallel wire windings **140**, **142**. If desired, however, a seam may be placed between every other pair of neighboring windings **140**, **142** or only select pairs of neighboring windings. In this regard, the subject disclosure is not per se limited to a particular type of stitch and, thus, may employ other conventional and unconventional stitch types, including chainstitches, lockstitches, overlock stitches, cover stitches, etc. As yet a further option, the windings **140**, **142** and threads **144**, **146** may be elongated along rectilinear paths, curvilinear paths, or any assorted combination of

geometric paths.

(22) To help retain the superposed wires in a tensioned state while concomitantly minimizing wire motion during the wire joining process, the superposed wire windings **140**, **142** may be stretched taut across a workpiece frame **200** (also referred to herein as “jig”) of FIG. **3**. In accordance with the illustrated example, the workpiece frame **200** is composed of multiple frame walls, namely first, second and third casing walls **202**, **204** and **206**, respectively, that collectively define an inner frame space **201** across which the superposed, unwoven wires are stretched to form a workpiece **132'**. The first and second casing walls **202**, **204**, which are generally straight and substantially colinear, may be connected to each other via the third wall **206**, which is shown having an elongated, arcuate shape. In at least some desired implementations, the casing walls **202**, **204**, **206** are integrally formed as a single-piece, unitary structure with the first and second casing walls **202**, **204** each projecting inward from a respective end of the third casing wall **206**. Clearly, the shape and size of the workpiece frame **200** of FIG. **3** is purely representative by nature, and is therefore non-limiting in scope.

(23) Spaced along the length of each casing wall **202**, **204**, **206** is a series of mechanical fastening features **208** (e.g., snap-fastener heads) for securing the workpiece frame **200** to a subjacent support surface, such as the assembly benchtop of a workstation table or a conveyor belt of a manufacturing system. Additionally, a series of cylindrical wire posts **210** projects generally orthogonally from the upper surface of each casing wall **202**, **204**, **206** for receiving the superposed wire workpiece **132'**. Like the mechanical fastening features **208**, the wire posts **210** are spaced from one another around the outer perimeter of the inner frame space **201**. The unwoven, superposed wires **140**, **142** are wound around these wire posts **210** to create the preliminary workpiece **132'**. Incidentally, manufacturing the engineered textile **132** may necessitate locating the superposed wires **140**, **142** in a tensioned, crisscrossed pattern on the workpiece frame **200** prior to joining of the wires **140**, **142**. Locating the superposed wires **140**, **142** may include manually or robotically anchoring then winding a first discrete wire in a first zigzag pattern around a first select set of the posts **210**, and subsequently anchoring then winding a second discrete wire in a second zigzag pattern around a second select set of the posts **210** such that the workpiece **132'** is stretched across the inner frame space **201**. The unwoven, overlapping wires **140**, **142** may be joined together at multiple predefined locations, e.g., via stitching, bonding, fusing and/or fastening the wires. For a footwear application, the anchoring points of the individual wires, the direction or directions of elongation of the individual wires, the points of overlap of the wires, and/or the locations of joining the wires may be data mapped to an intended user or users foot/feet to provide, for example, improved foot retention, comfort, performance, energy return, etc.

(24) Turning next to FIG. **4**, there is presented an automated manufacturing system **300** for constructing an engineered textile product, such as the engineered textile **132** surface of footwear upper **112** of FIG. **2**, from a workpiece composed of unwoven, superposed wires, such as workpiece **132'** of FIG. **3**. To remain pointed and succinct, only select components of the manufacturing system **300** have been shown and will be described in additional detail below. Nevertheless, the manufacturing systems and devices discussed herein may include numerous additional and alternative features, as well as other commercially available peripheral components, for example, to carry out the various protocols and algorithms of this disclosure. To this end, the automated manufacturing system **300** is portrayed in the Figures and described below as having a controller-automated, vision-guided robotic architecture; notwithstanding, the system **300** may take on other suitable architectures, including those using sensor-based automation and glide-track, carriage-borne precision movement.

(25) Manufacturing system **300** uses sensor-based and/or vision-guided stitching to automate the construction of an engineered textile having a desired shape and a set of desired functional characteristics. The representative architecture of FIG. **4** employs a movable end effector, such as a wall, ceiling or floor mounted robotic stitching cell **302**, that communicates, e.g., wired or

wirelessly, with a robot system controller **304** that governs operation of the cell **302**. The robotic stitching cell **302** includes a support frame **306** mounted to a distal end of an articulating robot arm **308**. Alternative embodiments may utilize a movable end effector composed of a support carriage that is slidably mounted for multidirectional movement on a slide track frame (not shown). As will be described in further detail hereinbelow, the robotic stitching cell **302** is designed to selectively complete one or more stitching operations along one or more seam joint regions of one or more workpieces. Movement of the articulating robot arm **308** may be provided by means of servomotors, linear and rotational transducers, pneumatic actuators, hydraulic actuators, or by any other type of logically applicable actuation mechanism. In the same vein, the robot arm **308** may have six degrees of freedom of motion, as shown, or have any other suitable number of degrees of freedom of motion.

(26) A processing head for joining superposed wires, such as stitching head **310**, is mounted via the support frame **306** to the articulating robot arm **308** above a telescoping benchtop table **312** of a manufacturing system workstation **314**. The processing head may take on various suitable formats, including a weld head for fusing the wires, an adhesive head for bonding the wires, a fastener head for mechanically joining the wires, etc. In accord with the illustrated example, the stitching head **310** includes a first (top) thread feeder **316** through which a metered length of a first (top) thread is selectively discharged. Mounted in opposing spaced relation to the first thread feeder **316** is a second (bottom) thread feeder, represented in FIG. 4 as a bobbin case **318**, through which a metered length of a second (bobbin) thread is selectively discharged. A sewing needle **320** is received and operatively retained by a motor-actuated needle receiver **322**. One or more integrated circuit (IC) processors **324** internal to robot system controller **304** execute stitch head control logic stored as a first control module **321** in resident memory device **326** to effectuate reciprocating motion (up-and-down translation in FIG. 4) of the sewing needle **320** via the needle receiver **322**. A shuttle hook **328** juxtaposed with the needle **320** and needle receiver **322** is operable to create a lockstitch between the bobbin thread fed from the bobbin case **318** and the top thread fed from the top thread feeder **316**.

(27) As indicated above, robot system controller **304** is constructed and programmed to automate, among other things, the movement and operation of the manufacturing system **300**. Control module, module, controller, control unit, electronic control unit, processor, and any permutations thereof may be defined to include any one or various combinations of one or more of logic circuits, Application Specific Integrated Circuit(s) (ASIC), electronic circuit(s), central processing unit(s) (e.g., microprocessor(s)), input/output circuit(s) and devices, appropriate signal conditioning and buffer circuitry, and other components to provide the described functionality, etc. Associated memory and storage (e.g., read only, programmable read only, random access, hard drive, tangible, etc.), shown schematically at **326** in FIG. 4, whether resident, remote or a combination of both, store processor-executable software, firmware programs, modules, routines, etc., which are collectively represented at **321**, **323** and **325**.

(28) Software, firmware, programs, instructions, routines, code, algorithms, and similar terms may be used interchangeably and synonymously to mean any processor-executable instruction sets, including calibrations and look-up tables. The system controller **304** may be designed with a set of control routines and logic executed to provide the desired functions. Control routines are executed, such as by a central processing unit, and are operable to monitor inputs from sensing devices and other networked control modules, and execute control and diagnostic routines to control operation of devices and actuators. Routines may be executed in real-time, continuously, systematically, sporadically and/or at regular intervals, for example, each 100 microseconds, 3.125, 6.25, 12.5, 25 and 100 milliseconds, etc., during ongoing use or operation of the system **300**.

(29) As shown in FIG. 4, the stitching head **310** carries a high-precision digital camera **330** (e.g., NAC MEMRECAM MX® Processing Optic) operable to capture, among other things, real-time digital images of a workpiece (e.g., workpiece **132'** of FIG. 3). This digital camera **330** operates as

a sensing device within a monitoring subsystem that is integrated into manufacturing system **300**, and may include an actuator-driven autofocus device and a multifocal module for controller operation of the autofocus device. The digital camera **330** senses one or more objects and generates feedback data, detects respective locations of select sites with respect to each object, and subsequently sends location image signals back to an image processing logic within a second memory-stored module **323**. The motorized autofocus device provides systematic precision focus upon the objects and displacement between designated sites, e.g., by being adjusted to be closer to and farther from each object/site. Image data generated by the digital camera **330** and processed through the image processing module **323** is passed to the stitch head control logic stored in the first control module **321** and robot control logic stored as a third control module **325** in resident memory device **326** to automate lockstitching together the unwoven, superposed wires of an engineered textile workpiece.

(30) With reference next to the flow chart of FIG. 5, an improved method or control strategy for automating operation of a manufacturing system, such as manufacturing system **300** of FIG. 4, to create an engineered textile product, such as engineered textile **132** of FIG. 2, from a workpiece, such as workpiece **132'** of FIG. 3, is generally described at **400** in accordance with aspects of the present disclosure. Some or all of the operations illustrated in FIG. 5 and described in further detail below may be representative of an algorithm that corresponds to processor-executable instructions that may be stored, for example, in main or auxiliary or remote memory, and executed, for example, by an on-board or off-board controller, processing unit, control logic circuit, or other module or device or network of modules/devices, to perform any or all of the above or below described functions associated with the disclosed concepts. It should be recognized that the order of execution of the illustrated operation blocks may be changed, additional blocks may be added, and some of the blocks described may be modified, combined, or eliminated.

(31) Method **400** begins at terminal block **401** of FIG. 5 with processor-executable instructions for a programmable controller or control module or similarly suitable processor to call up an initialization procedure for a closed-loop control sequence with real-time stitch head guidance and adjustment during an automated stitching operation. Block **401** may initialize in response to a user prompt from a system operator or technician of the robotic stitching cell **302**, or responsive to a broadcast prompt signal from a backend server-class computer or middleware computing node tasked with governing operation of a robotic cell, collection of robot cells, or a manufacturing facility incorporating therein one or more robot cells. To carry out this protocol, a control system or any combination of one or more subsystems may be operable to receive, process, and synthesize pertinent information and inputs, and execute control logic and algorithms to regulate various subsystems and/or subsystem components to achieve desired control targets. As part of initiating the method **400** at terminal block **401**, an initial system setup may be carried out on the robotic stitching cell **302** through a suitable human machine interface (HMI), including powering on the various system components, calibrating an origin position, and identifying respective current locations of the stitch head and workpiece relative to this calibrated origin position.

(32) Method **400** of FIG. 5 advances from terminal block **401** to input/output block **403** with processor-executable instructions for a control device, such as robot system controller **304**, to exchange data with an image capture device, such as high-precision digital camera **330** of FIG. 4. Data generated by the image capture device may be indicative of a real-time captured image or set of images of a plan view and/or perspective view of a workpiece. At predefined process block **405**, instructions are provided for the control device to analyze the captured workpiece image(s) and, through this evaluation, locate the interwire gaps within the quadrangles defined by the crisscrossed, superposed wires of the workpiece. By way of non-limiting example, robot system controller **304** of FIG. 4 may employ image processing module **323** stored within memory device **326** to: (1) process and filter the captured image of the workpiece (e.g., focusing, zooming, sharpening, edge detection, object detection, etc.); (2) scan the image to identify respective sets of

two, three, or four intersecting points of the superposed wires that collectively define, in whole or in part, each quadrangle; (3) for each respective set of intersecting points, derive a center of a respective diagonal line segment that connects an opposing pair of the intersecting points (e.g., a geometric “diagonal” connecting opposite vertices of a polygon); and (4) designating the centers of the diagonal line segments of the intersecting point sets as the interwire gaps.

(33) Accurate gap identification and location for provisioning vision-guided precision stitching may be achieved through various supplementary or alternative techniques to those described in the preceding section. For instance, robot system controller **304** may employ image processing module **323** to: (1) process and filter the captured workpiece image(s); (2) from these processed and filtered image(s), approximate a centerline or lateral edge line for each superposed wire; (3) construct the superposed wire quadrangles from these estimated centerlines/edge lines; and (4) designate a central region within each quadrangle between the estimated centerlines as one of the gaps.

Optionally, robot system controller **304** may employ image processing module **323** to: (1) process and filter the captured image of the workpiece; (2) evaluate the processed and filtered workpiece image(s) to derive at least two wire intersecting points that define at least two respective corners on a common edge of each quadrangle; (3) determine, for each quadrangle, a central region defined at: (i) a calibrated angle from a line segment connecting the two respective corners, and (ii) a calibrated distance from one of the respective corners; and (4) categorize these central regions of the quadrangles as the interwire gaps.

(34) Rather than identifying gap locations for each workpiece on an individualized basis, predefined process block **405** may provide product-specific routing instructions for processing a succession of workpieces intended to make multiples of a particular product. For instance, robot system controller **304** of FIG. **4** may lookup or otherwise retrieve from resident memory device **326** a customized set of pre-defined path plan instructions that have been mapped out for the stitching head to insert a succession of stitches within the gaps between the superposed wires. The foregoing path plan data may comprise, among other things, a stitch head start position or “origin,” a stitch head end position or “destination,” and a calibrated stitching route for moving the stitching head from the origin to the destination in order to insert the stitches at designated points along the route. Other path plan data may include stitch head speed, fore-aft pitch angle and rate, lateral pitch angle and rate, etc.

(35) Once the system identifies the desired gap locations into which stitches will be inserted for mechanically interconnecting the superposed wires of the workpiece, method **400** proceeds to process block **407** and initiates automated stitching. To do so, robot system controller **304** may transmit one or more electronic command signals to the articulating robot arm **308** to sequentially move the stitching head **310** across the exposed face of the workpiece **132'** and precisely align the sewing needle **320** with each of the quadrangle's internal gaps. The vision-based guidance system may be employed to ensure accurate alignment of the needle receiver **322** and shuttle hook **328** with respect to the interwire gaps prior to inserting a stitch. Process block **407** may also provide instructions that direct the robot system controller **304** to transmit one or more electronic command signals to the stitching head **310** to insert a succession of stitches within the gaps between the superposed wires.

(36) Precision control of the automated stitching process may be further enabled through real-time position tracking of the stitching head **310**. One or more optical position sensors **332** may be mounted at discrete locations of the robotic stitching cell **302** to determine real-time positions of the stitching head **310**, e.g., relative to a calibrated origin position. Robot system controller **304** receives from the position sensor(s) **332** one or more sensor signals that are indicative of the real-time positions of the stitching head **310**. If so desired, the system controller **304** may determine, from the received sensor signal(s) and the captured image(s) of the workpiece, an estimated distance between each real-time position of the stitching head **310** and a respective location of the next gap adjacent the stitching head's current position. Automated movement of the articulating

robot arm **308** may include estimating a desired trajectory for moving the stitching head **310** from its current position to the location of the next gap based on the corresponding estimated distance between the stitching head and next adjacent gap. From the received sensor signal(s) and the captured workpiece image(s), the robot system controller **304** may also locate—one-at-a-time in real-time—the next adjacent gap that is closest to the current real-time position of the stitching head **310**.

(37) It may be desirable, for any of the above implementations, to make real-time adjustments to the stitching route parameters in order to accommodate part-to-part variations, manufacturing tolerances, inadvertent wire displacement, etc. This may include process block **409** providing instructions for the system controller to pull a trajectory trace of the stitch route, identifying start and end positions within a captured image of the workpiece, and superimposing the trace of the stitch route onto the captured image of the workpiece with the origin overlapping the start position and the destination overlapping the end position. After superimposing the trace onto the captured image, the system controller identifies one or more part-specific calibrated alignment points on the stitch route, and determines if there is any displacement between each calibrated alignment point and a corresponding alignment location in the image of the workpiece. If so, the system responsively determines and implements a respective trace correction to offset each respective displacement. Once the foregoing operations are completed, the method **400** of FIG. 5 may advance to terminal block **411** and terminate, or may loop back to terminal block **401** and run in a continuous loop.

(38) Aspects of this disclosure may be implemented, in some embodiments, through a computer-executable program of instructions, such as program modules, generally referred to as software applications or application programs executed by any of a controller or the controller variations described herein. Software may include, in non-limiting examples, routines, programs, objects, components, and data structures that perform particular tasks or implement particular data types. The software may form an interface to allow a computer to react according to a source of input. The software may also cooperate with other code segments to initiate a variety of tasks in response to data received in conjunction with the source of the received data. The software may be stored on any of a variety of memory media, such as CD-ROM, magnetic disk, and semiconductor memory (e.g., various types of RAM or ROM).

(39) Moreover, aspects of the present disclosure may be practiced with a variety of computer-system and computer-network configurations, including multiprocessor systems, microprocessor-based or programmable-consumer electronics, minicomputers, mainframe computers, and the like. In addition, aspects of the present disclosure may be practiced in distributed-computing environments where tasks are performed by resident and remote-processing devices that are linked through a communications network. In a distributed-computing environment, program modules may be located in both local and remote computer-storage media including memory storage devices. Aspects of the present disclosure may therefore be implemented in connection with various hardware, software or a combination thereof, in a computer system or other processing system.

(40) Any of the methods described herein may include machine readable instructions for execution by: (a) a processor, (b) a controller, and/or (c) any other suitable processing device. Any algorithm, software, control logic, protocol or method disclosed herein may be embodied as software stored on a tangible medium such as, for example, a flash memory, solid-state drive (SSD) memory, hard-disc drive (HDD) memory, a CD-ROM, a digital versatile disk (DVD), or other memory devices. The entire algorithm, control logic, protocol, or method, and/or parts thereof, may alternatively be executed by a device other than a controller and/or embodied in firmware or dedicated hardware in an available manner (e.g., implemented by an application specific integrated circuit (ASIC), a programmable logic device (PLD), a field programmable logic device (FPLD), discrete logic, etc.). Further, although specific algorithms are described with reference to flowcharts depicted herein,

many other methods for implementing the example machine-readable instructions may alternatively be used.

(41) Aspects of the present disclosure have been described in detail with reference to the illustrated embodiments; those skilled in the art will recognize, however, that many modifications may be made thereto without departing from the scope of the present disclosure. The present disclosure is not limited to the precise construction and compositions disclosed herein; any and all modifications, changes, and variations apparent from the foregoing descriptions are within the scope of the disclosure as defined by the appended claims. Moreover, the present concepts expressly include any and all combinations and subcombinations of the preceding elements and features. Additional features may be reflected in the following clauses: Clause 1: an automated manufacturing system for constructing an engineered textile from a workpiece composed of superposed wires, the manufacturing system comprising: a movable end effector; a processing head mounted to the movable end effector; and a system controller operatively connected to the movable end effector and the processing head, the system controller being programmed to: determine a plurality of predefined joint locations of the superposed wires; command the movable end effector to sequentially move the processing head to align with each of the predefined joint locations; and command the processing head to join the superposed wires at the predefined joint locations to form the engineered textile. Clause 2: a manufacturing system of clause 1, wherein the processing head includes a stitching head with a thread feeder and a sewing needle cooperatively configured to generate stitches. Clause 3: a manufacturing system of clause 2, further comprising an image capture device mounted to the movable end effector and configured to capture an image of the workpiece and output data indicative thereof. Clause 4: a manufacturing system of clause 3, wherein the system controller is further programmed to: receive, from the image capture device, the data indicative of the captured image of the workpiece, wherein determining the plurality of predefined joint locations includes locating, from the captured image of the workpiece, multiple gaps each defined between a quadrangle of the superposed wires, and wherein commanding the processing head to join the superposed wires includes commanding the stitching head to insert a succession of stitches within the gaps between the superposed wires. Clause 5: a manufacturing system of clause 4, wherein the system controller is further programmed to: identify, within the captured image of the workpiece, respective sets of intersecting points of the superposed wires defining the quadrangles; and determine, within each of the respective sets, a center of a respective diagonal line segment connecting an opposing pair of the intersecting points, wherein locating the gaps includes designating the center of the diagonal line segment of each of the sets of intersecting points as one of the gaps. Clause 6: a manufacturing system of clause 4, wherein the system controller is further programmed to: identify, within the captured image of the workpiece, an estimated centerline for each of the superposed wires; and construct the quadrangles of the superposed wires from the estimated centerlines, wherein locating the gaps includes designating a central region within each of the quadrangles between the estimated centerlines as one of the gaps. Clause 7: a manufacturing system of clause 4, wherein the system controller is further programmed to: identify, within the captured image of the workpiece, two intersecting points of the superposed wires defining two respective corners for each of the quadrangles; and determine, for each of the quadrangles, a central region at a calibrated angle from a line segment connecting the two respective corners and at a calibrated distance from one of the respective corners, wherein locating the gaps includes designating the central region of each of the quadrangles as one of the gaps. Clause 8: a manufacturing system of any one of clauses 1 to 8, wherein the system controller is further programmed to determine path plan data for moving the processing head to join the superposed wires at the predefined joint locations, the path plan data including an origin, a destination, and a joint route for traversing the processing head from the origin to the destination. Clause 9: a manufacturing system of clause 8, wherein the system controller is further programmed to: generate a trace of the joint route; determine a start position and an end position within the

captured image of the workpiece; and superimpose the trace of the joint route onto the captured image of the workpiece with the origin overlapping the start position and the destination overlapping the end position. Clause 10: a manufacturing system of clause 9, wherein the system controller is further programmed to: determine a plurality of calibrated alignment points on the stitch route; determine a respective displacement, if any, between each of the calibrated alignment points and a respective alignment location in the image of the workpiece; and determine a respective trace correction to offset each of the respective displacements. Clause 11: a manufacturing system of any one of clauses 1 to 10, further comprising a workpiece frame configured to retain the superposed wires in a tensioned, crisscrossed pattern. Clause 12: a manufacturing system of clause 11, wherein the workpiece frame includes a plurality of adjoining casing walls defining an inner frame space across which the workpiece is stretched, and a series of posts projecting from the casing walls and spaced from one another along the perimeter of the inner frame space, the wires being wound around the posts. Clause 13: a manufacturing system of any one of clauses 1 to 12, further comprising a position sensor configured to determine real-time positions of the processing head relative to a calibrated origin position and output sensor signals indicative thereof. Clause 14: a manufacturing system of clause 13, wherein the system controller is further programmed to: receive, from the position sensor, the sensor signals indicative of the real-time positions of the processing head; and determine, from the received sensor signals and a captured image of the workpiece, an estimated distance between each of the real-time positions of the processing head and a next adjacent one of the joint locations; and estimate a plurality of desired trajectories each based on the estimated distance between the real-time position of the processing head and the respective next adjacent one of the joint locations. Clause 15: a manufacturing system of clause 14, wherein the system controller is further programmed to determine, one-at-a-time in real-time from the received sensor signals and the captured image of the workpiece, the respective next adjacent one of the joint locations closest to each of the real-time positions of the stitching head. Clause 16: a manufacturing system of any one of clauses 1 to 15, wherein the movable end effector includes a support frame attached to a robot arm. Clause 17: a manufacturing system of any one of clauses 1 to 15, wherein the movable end effector includes a support carriage attached to a slide track frame. Clause 18: an article of footwear for a foot of a user, the article of footwear comprising: a sole structure configured to support thereon the foot of the user; and an upper attached to the sole structure and configured to attach to the foot of the user, the upper including an upper segment fabricated from an engineered textile, the engineered textile including: a first set of mutually parallel wire windings elongated in a first direction; a second set of mutually parallel wire windings elongated in a second direction distinct from the first direction, the second set of wire windings abutting the first set of wire windings in an unwoven, intercrossed pattern, wherein the abutting first and second sets of wire windings are joined at a plurality of predefined joint locations. Clause 19: an article of footwear of clause 18, wherein the unwoven, intercrossed pattern defines a plurality of quadrangles with central gaps, and wherein the article of footwear further comprises first and second threads elongated in a third direction parallel with respect to the first direction, wherein the first and second threads are lockstitched together in the central gaps between the intercrossed wires. Clause 20: a non-transitory, computer-readable medium storing instructions for execution by one or more processors of a system controller of an automated manufacturing system, the instructions causing the automated manufacturing system to perform operations comprising: receiving, from an image capture device mounted to a movable end effector, data indicative of a captured image of a workpiece, the workpiece being composed of multiple unwoven, superposed wires, the movable end effector having mounted thereto a stitching head with a thread feeder and a sewing needle cooperatively configured to generate stitches; locating, from the captured image of the workpiece, multiple gaps each defined between a quadrangle of the superposed wires; command the movable end effector to sequentially move the stitching head and thereby align the sewing needle with each of the gaps; and command the

stitching head to insert a succession of stitches within the gaps between the superposed wires. Clause 21: a method of operating an automated manufacturing system, the method comprising: receiving a workpiece composed of superposed wires; determining, via a system controller, a plurality of predefined joint locations for the superposed wires; commanding, via the system controller, a movable end effector to sequentially move a processing head to thereby align the processing head with each of the predefined joint locations; and commanding, via the system controller, the processing head to join the superposed wires at the predefined joint locations to form an engineered textile. Clause 22: a method of clause 21, further comprising receiving, via the system controller from an image capture device, data indicative of a captured image of the workpiece. Clause 23: a method of clause 21 or clause 22, wherein the processing head includes a stitching head with a thread feeder and a sewing needle cooperatively configured to generate stitches. Clause 24: a method of clause 23, wherein the stitching head further includes a needle receiver operable to reciprocally translate the sewing needle, a bobbin case operable to feed bobbin thread, and a shuttle hook operable to create a lockstitch between the bobbin thread and a top thread fed from the thread feeder. Clause 25: a method of any one of clauses 22 to 24, wherein determining the predefined joint locations for the superposed wires includes locating, via the system controller from the captured image of the workpiece, multiple gaps each defined between a quadrangle of the superposed wires, and wherein commanding the processing head to join the superposed wires at the predefined joint locations includes commanding, via the system controller, the stitching head to insert a succession of stitches within the gaps between the superposed wires. Clause 26: a method of clause 24, further comprising: identifying, within the captured image of the workpiece, respective sets of intersecting points of the superposed wires defining the quadrangles; and determining, within each of the respective sets, a center of a respective diagonal line segment connecting an opposing pair of the intersecting points, wherein locating the gaps includes designating the center of the diagonal line segment of each of the sets of intersecting points as one of the gaps. Clause 27: a method of clause 24, further comprising: identifying, within the captured image of the workpiece, an estimated centerline for each of the superposed wires; and constructing the quadrangles of the superposed wires from the estimated centerlines, wherein locating the gaps includes designating a central region within each of the quadrangles between the estimated centerlines as one of the gaps. Clause 28: a method of clause 24, further comprising: identifying, within the captured image of the workpiece, two intersecting points of the superposed wires defining two respective corners for each of the quadrangles; and determining, for each of the quadrangles, a central region at a calibrated angle from a line segment connecting the two respective corners and a calibrated distance from one of the respective corners, wherein locating the gaps includes designating the central region of each of the quadrangles as one of the gaps. Clause 29: a method of any one of clauses 21 to 29, further comprising determining path plan data for moving the processing head to join the superposed wires at the predefined joint locations, the path plan data including an origin, a destination, and a route for traversing the processing head from the origin to the destination. Clause 30: a method of clause 29, further comprising: generating a trace of the route; determining a start position and an end position within the captured image of the workpiece; and superimposing the trace of the route onto the captured image of the workpiece with the origin overlapping the start position and the destination overlapping the end position. Clause 31: a method of clause 30, further comprising: identifying a plurality of calibrated alignment points on the route; determining a respective displacement, if any, between each of the calibrated alignment points and a respective alignment location in the image of the workpiece; and determining a respective trace correction to offset each of the respective displacements. Clause 32: a method of any one of clauses 21 to 31, wherein the automated manufacturing system further includes a workpiece frame, the method further comprising locating the superposed wires in a tensioned, crisscrossed pattern in the workpiece frame. Clause 33: a method of clause 32, wherein the workpiece frame includes a plurality of adjoining casing walls defining an inner frame space,

and a series of posts projecting from the casing walls and spaced from one another along the perimeter of the inner frame space, and wherein locating the superposed wires includes winding the wires around the posts such that the workpiece is stretched across the inner frame space. Clause 34: a method of any one of clauses 21 to 33, wherein the automated manufacturing system further includes a position sensor, the method further comprising receiving, via the system controller from the position sensor, sensor signals indicative of real-time positions of the processing head relative to a calibrated origin position. Clause 35: a method of clause 34, further comprising determining, from the received sensor signals and the captured image of the workpiece, an estimated distance between each of the real-time positions of the processing head and a next adjacent one of the joint locations, wherein commanding the movable end effector to move the processing head includes estimating a plurality of desired trajectories each based on the estimated distance between the real-time position of the processing head and the respective next adjacent one of the joint locations. Clause 36: a method of clause 35, further comprising determining, one-at-a-time in real-time from the received sensor signals and the captured image of the workpiece, the respective next adjacent one of the joint locations closest to each of the real-time positions of the processing head. Clause 37: a method of clause 21, wherein the movable end effector includes a support frame attached to a robot arm or a support carriage attached to a slide track frame.

Claims

1. An article of footwear, comprising: a sole structure configured to support thereon a foot of a user; and an upper attached to the sole structure and configured to attach to the foot of the user, the upper including an upper segment fabricated from an engineered textile, the engineered textile including: a first set of wire windings substantially parallel to one another and elongated in a first direction; a second set of wire windings substantially parallel to one another and elongated in a second direction distinct from the first direction, the second set of wire windings abutting the first set of wire windings in an unwoven, intercrossed pattern; a first plurality of substantially linear stitch seams located between the first set of wire windings, contacting the second set of wire windings, and binding the second set of wire windings to one another; and a second plurality of substantially linear stitch seams located between the second set of wire windings, contacting the first set of wire windings, and binding the first set of wire windings to one another, wherein each substantially linear stitch seam of the first and second pluralities of substantially linear stitch seams includes first and second threads codirectional with each other and defining a plurality of joints located at predefined joint locations within the intercrossed pattern and joining together the first and second sets of wire windings.
2. The article of footwear of claim 1, wherein each of the joints includes a respective lockstitch of the first thread and the second thread.
3. The article of footwear of claim 2, wherein diameters of the first and second sets of wire windings are at least 75% larger than diameters of the first and second threads.
4. The article of footwear of claim 2, wherein the first thread lays on top of and abuts the first and second sets of wire windings in an unwoven, intercrossed pattern.
5. The article of footwear of claim 1, wherein first portions of the first and second threads define the first plurality of substantially linear stitch seams and are elongated in a third direction parallel with respect to the first direction.
6. The article of footwear of claim 5, wherein second portions of the first and second threads define the second plurality of substantially linear stitch seams and are elongated in a fourth direction substantially perpendicular to the first and third directions.
7. The article of footwear of claim 1, wherein the first and second sets of wire windings have a first diameter, the first thread has a second diameter less than the first diameter, and the second thread has a third diameter less than the first and second diameters.

8. The article of footwear of claim 1, wherein the unwoven, intercrossed pattern defines a plurality of quadrangles with central gaps, and wherein each of the joints is located in a respective one of the central gaps.
9. The article of footwear of claim 8, wherein each of the quadrangles is defined by a respective set of intersecting points at which the first set of wire windings intercrosses the second set of wire windings, and wherein each of the joints is located at a respective center of a diagonal line segment connecting an opposing pair of the intersecting points.
10. The article of footwear of claim 1, wherein the upper segment includes a vamp portion and/or a toe box portion of the upper.
11. The article of footwear of claim 1, wherein the first and second sets of wire windings are retained in a tensioned state in the unwoven, intercrossed pattern.
12. The article of footwear of claim 1, wherein the first set of wire windings sits on top of and is moveable with respect to the second set of wire windings.
13. The article of footwear of claim 12, wherein the first set of wire windings is translatable with respect to the second set of wire windings in the first direction, and the second set of wire windings is translatable with respect to the first set of wire windings in the second direction.
14. The article of footwear of claim 1, wherein the first and second sets of wire windings are formed from first and second wire strands, respectively, each formed from an aliphatic or semi-aromatic polyamide material, an aromatic polyester material, an aramid material, a polypropylene material, and/or a polyethylene material.
15. An article of footwear, comprising: a sole structure configured to support thereon a foot of a user; and an upper attached to the sole structure and configured to attach to the foot of the user, the upper including an upper segment fabricated from an engineered textile, the engineered textile including: a first set of wire windings parallel to one another and elongated in a first direction; a second set of wire windings parallel to one another and elongated in a second direction perpendicular to the first direction, the second set of wire windings abutting the first set of wire windings in an unwoven, intercrossed pattern; a first plurality of substantially linear stitch seams located between the first set of wire windings, contacting the second set of wire windings, and binding the second set of wire windings to one another; and a second plurality of substantially linear stitch seams located between the second set of wire windings, contacting the first set of wire windings, and binding the first set of wire windings to one another, wherein each substantially linear stitch seam of the first and second pluralities of substantially linear stitch seams includes first and second threads codirectional with each other and defining a plurality of joints located at predefined joint locations within the intercrossed pattern and joining together the first and second sets of wire windings, wherein the first direction of the first set of wire windings is substantially perpendicular to the second direction of the second set of wire windings.
16. The article of footwear of claim 15, wherein each of the joints includes a respective lockstitch of the first thread and the second thread.
17. The article of footwear of claim 15, wherein each of the first and second threads in the first plurality of substantially linear stitch seams extends substantially parallel to the first direction and is interposed between a respective pair of wire windings in the first set of wire windings.
18. The article of footwear of claim 15, wherein the second plurality of substantially linear stitch seams is substantially perpendicular to the first plurality of substantially linear stitch seams.
19. The article of footwear of claim 15, wherein each of the first and second threads in the second plurality of substantially linear stitch seams extends substantially parallel to the second direction and is interleaved with a respective pair of wire windings in the second set of wire windings.
20. The article of footwear of claim 15, wherein the unwoven, intercrossed pattern includes the first and second sets of wire windings arranged in a square array of substantially perpendicular rows and columns.
21. An article of footwear, comprising: a sole structure configured to support thereon a foot of a

user; and an upper attached to the sole structure and configured to attach to the foot of the user, the upper including an upper segment fabricated from an engineered textile, the engineered textile including: a first set of wire windings substantially parallel to one another and elongated in a first direction; a second set of wire windings substantially parallel to one another and elongated in a second direction substantially perpendicular to the first direction, the second set of wire windings abutting the first set of wire windings in an unwoven, intercrossed pattern; a first plurality of substantially linear stitch seams located between the first set of wire windings, contacting the second set of wire windings, and binding the second set of wire windings to one another; and a second plurality of substantially linear stitch seams located between the second set of wire windings, contacting the first set of wire windings, and binding the first set of wire windings to one another, wherein the first plurality of substantially linear stitch seams is parallel with the first set of wire windings and perpendicular to the second set of wire windings, and the second plurality of substantially linear stitch seams is parallel with the second set of wire windings and perpendicular to the first set of wire windings, and wherein each substantially linear stitch seam of the first and second pluralities of substantially linear stitch seams is formed by first and second lockstitched threads codirectional with each other and defining a plurality of joints joining together the first and second sets of wire windings.
